TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

THAPATHALI CAMPUS

A PROJECT PROPOSAL

ON

EXPLAINABLE WIND-AWARE SPATIO-TEMPORAL GRAPH NEURAL NETWORK FOR AIR QUALITY FORECASTING AND MITIGATION ANALYSIS

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ABSTRACT

Ambient air pollution is a major environmental and public-health concern in urban areas where pollutant concentration varies across space, time and season. Kathmandu Valley is affected by road traffic, dust resuspension, construction activity, brick kilns, forest-fire episodes, valley topography and changing meteorological conditions. Traditional station-wise forecasting models often ignore wind-driven pollutant movement between monitoring stations and usually provide numerical results without explaining why a forecast is high or low. This proposal presents an explainable wind-aware Spatio-Temporal Graph Neural Network (ST-GNN) for short-term PM2.5 forecasting and mitigation analysis. Monitoring stations are represented as graph nodes, while dynamic edges are constructed from geographical distance, historical PM2.5 correlation, wind speed and wind-direction alignment. The primary model is a Graph Attention Network with Gated Recurrent Unit (GAT-GRU), where GAT captures spatial influence and GRU captures temporal patterns. The task is formulated as regression for hourly PM2.5 forecasting at 6-hour horizon, while Air Quality Index categories and pollution-peak labels are derived from the predicted concentration for interpretation. The system includes preprocessing rules for null values, sensor faults and real pollution spikes, along with explainability using attention weights, feature attribution and temporal sensitivity. The final output will be a dashboard showing station-level forecasts, AQI risk classes, historical-versus-predicted graphs, spatial pollution maps, feature-importance plots. The project is intended as a research prototype and decision-support tool, not as a replacement for official air-quality warning systems.

Keywords*—Air quality forecasting, $PM_{2.5}$, wind-aware graph, spatio-temporal graph neural network, GAT-GRU, explainable AI, AQI, Kathmandu Valley.*

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1 INTRODUCTION

1.1 Background

Air pollution is one of the most serious environmental issues affecting public health, urban planning, and quality of life. Pollutants such as $PM_{2.5}$, PM_{10}, NO_2, O_3, CO, and SO_2 can vary significantly over time and across locations. These variations are influenced by traffic, industrial activity, weather, wind direction, wind speed, temperature, humidity, and regional pollutant transport.

Air quality forecasting is therefore a spatio-temporal problem. The pollutant concentration at one monitoring station depends not only on its previous readings but also on neighboring stations, meteorological conditions, and the transport of pollutants from surrounding areas. Machine learning and deep learning models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) can learn temporal patterns, but they may not fully capture spatial relationships between monitoring stations when each station is treated independently.

Graph Neural Networks (GNNs) are suitable for learning from relational data. In the proposed project, the air quality monitoring network is represented as a graph. Each monitoring station becomes a node, and the relationship between two stations becomes an edge. By combining graph-based spatial learning with recurrent temporal learning, the system can model both station-to-station influence and pollutant variation over time.

1.2 Motivation

Traditional station-wise forecasting can be insufficient because pollution does not remain fixed around one station. Pollutants can travel from one area to another due to wind and atmospheric movement. A wind-aware graph-based model can provide better insight into how pollution may spread across different monitoring stations and can adapt edge influence when wind conditions change.

Another motivation is explainability. Forecasting models should not only provide predicted pollutant values but also explain why a certain area is expected to have high pollution. By analyzing attention weights, feature importance, wind direction, and neighboring station influence, the proposed system aims to provide human-understandable explanations for air quality predictions.

1.3 Problem Definition

Air quality forecasting is a challenging spatio-temporal problem because pollutant concentration at a monitoring station depends not only on its past readings, but also on surrounding stations, meteorological conditions, and wind-driven pollutant movement.

Many existing forecasting approaches either treat monitoring stations independently or use fixed spatial relationships, which may not accurately represent changing pollutant dispersion caused by variations in wind direction and wind speed.

Furthermore, several forecasting models provide numerical predictions without clearly explaining which neighboring stations, weather factors, or previous time steps influenced the forecast. This lack of explainability limits the usefulness of such models for environmental monitoring and decision-support. Existing systems also rarely provide scenario-based analysis to estimate how changes in pollution levels at selected locations may affect surrounding areas.

Therefore, the problem addressed in this project is to design and develop an explainable wind-aware spatio-temporal forecasting framework that can model dynamic relationships between air quality monitoring stations, predict future pollutant concentrations, identify important influencing factors, and support what-if scenario analysis for mitigation planning.

1.4 Objectives

The objectives of the project are:

1. To develop an explainable wind-aware Spatio-Temporal Graph Neural Network (ST-GNN) model for air quality forecasting using pollutant and meteorological data.
2. To design an interactive dashboard that visualizes predicted air quality, station-level risk, explanation of model predictions, and what-if mitigation scenarios.

1.5 Scope and Applications

The scope of this project includes air quality forecasting using data from public monitoring stations and meteorological sources. The system will focus on pollutants such as $PM_{2.5}$, PM_{10}, NO_2, and O_3, depending on dataset availability. The forecasting horizon may include short-term predictions such as 1 hour, 6 hours, 12 hours, and 24 hours.

The system will be developed as a research prototype and decision-support tool. It will not replace official government air quality warning systems. The applications of the project include air quality forecasting, environmental monitoring, pollution risk visualization, urban environmental planning support, station-level pollution influence analysis, public awareness dashboards, and what-if mitigation scenario analysis.

2 LITERATURE REVIEW

2.1 Air Quality Forecasting and the Need for Spatio-Temporal Modeling

Air pollution is a serious environmental and public-health issue because pollutant concentration does not remain constant across location or time. Pollutants such as $PM_{2.5}$, PM_{10}, NO_2, and O_3 are affected by many factors, including traffic emissions, industrial activities, dust, temperature, humidity, atmospheric pressure, wind speed, and wind direction. Because of this, forecasting air quality is not only a time-series problem but also a spatial problem. The pollution level at one monitoring station may depend on its own past readings, but it may also be influenced by nearby or upwind stations.

Traditional forecasting methods such as statistical models, regression-based models, support vector machines, random forest, LSTM, and GRU have been widely used for environmental prediction. These methods are useful as baseline models because they can learn historical trends and short-term temporal patterns. However, many of them treat each monitoring station independently. In reality, monitoring stations are connected through atmospheric movement and pollutant transport. For example, if pollution is high in one area and wind carries polluted air toward another area, the second area may experience increased pollution after some time. This kind of relationship cannot be fully captured by station-wise models alone.

Recent air quality forecasting studies have therefore moved toward spatio-temporal modeling. In this approach, the model learns both temporal dependency and spatial dependency. Temporal dependency means how pollution changes over hours or days, while spatial dependency means how one location may influence another. This idea is important for the proposed project because the system aims to forecast pollutant levels by combining historical pollution data, meteorological data, and relationships between monitoring stations.

2.2 Graph Neural Networks for Station-Based Air Quality Forecasting

GNNs provide a suitable framework for modeling air quality monitoring networks. In a graph-based representation, each air quality monitoring station can be considered as a node, and the relationship between two stations can be represented as an edge. This allows the model to learn how stations interact with each other rather than treating every station as completely independent.

The Graph Convolutional Network proposed by Kipf and Welling introduced an efficient method for learning from graph-structured data by aggregating information from neighboring nodes [1]. In the context of air quality forecasting, this means that a monitoring station can learn from its connected neighboring stations. This is useful because air pollution at one station may be affected by pollution conditions in surrounding areas.

However, all neighboring stations do not contribute equally. Some stations may have stronger influence because they are closer, more polluted, or located in the direction from which wind is blowing. Graph Attention Network, proposed by Velickovic et al., addresses this issue by allowing the model to assign different attention weights to different neighboring nodes [2]. This makes graph attention useful for air quality forecasting because the model can learn which neighboring station is more important for a particular prediction.

These graph-based approaches provide the foundation for the proposed project. Graph convolution can be used as a baseline spatial model, while graph attention can be used to learn unequal station influence. When combined with temporal models such as GRU or LSTM, these methods can support spatio-temporal air quality forecasting.

2.3 Spatio-Temporal GNNs and Domain Knowledge in Air Quality Prediction

Several recent studies have shown that air quality forecasting improves when graph learning is combined with domain knowledge and temporal modeling. Wang et al. proposed PM2.5-GNN, a domain-knowledge enhanced graph neural network for $PM_{2.5}$ forecasting [3]. Their work highlights that $PM_{2.5}$ concentration is influenced by complex environmental factors and that graph-based models can capture relationships between locations more effectively than simple time-series models. This supports the idea that air quality forecasting should include not only historical pollutant values but also spatial and meteorological relationships.

Pandey et al. proposed a spatial-diffusion guided encoder-decoder architecture for $PM_{2.5}$ forecasting [4]. Their model combines temporal learning with graph-based spatial diffusion, showing that $PM_{2.5}$ forecasting benefits from modeling how pollution spreads across monitoring locations. This is closely related to the proposed project because the planned system also combines graph-based spatial learning with recurrent temporal learning.

Panja et al. proposed E-STGCN, an extreme spatio-temporal graph convolutional network for air quality forecasting [5]. Their work focuses on high-pollution or extreme pollution events, which are important in public-health decision-making. Although the proposed project does not directly focus on extreme value theory, this study is useful because it shows that spatio-temporal graph models can be extended for risk-aware air quality forecasting.

These studies confirm that spatio-temporal GNNs are a strong direction for air quality prediction. However, a major limitation in many graph-based forecasting systems is that the graph structure is often fixed. A fixed graph may be created using distance or historical correlation, but real pollutant movement changes with wind and weather. This creates the need for dynamic and wind-aware graph construction.

2.4 Wind-Aware and Dynamic Graph Construction

The main technical idea of the proposed project is that the relationship between two air quality stations should not always remain fixed. In real atmospheric conditions, wind direction and wind speed can change throughout the day. If wind flows from one station toward another, the first station may have stronger influence on the second station. If the wind changes direction, the influence may also change. Therefore, a graph used for air quality forecasting should ideally be dynamic and wind-aware.

HighAir, proposed by Chen et al., is an important reference in this direction because it uses a hierarchical graph neural network for air quality forecasting and dynamically adjusts graph edge weights based on wind direction [6]. This directly supports the proposed project's idea that wind direction should influence station-to-station relationships.

Xu et al. proposed a Dynamic Graph Neural Network with Adaptive Edge Attributes for air quality prediction [7]. Their work argues that predefined graph structures may not always represent real relationships between monitoring sites. Instead, the model should learn or adjust edge attributes dynamically. This is important for the proposed system because the planned graph will not depend only on fixed distance. It will combine distance, historical pollutant correlation, wind direction, and wind speed to represent changing station influence.

The idea of dynamic graph construction is also connected to the physical behavior of pollutants. Air pollutants spread through diffusion and can be transported by wind through advection. AirPhyNet, proposed by Hettige et al., uses physics-guided neural networks for air quality prediction and incorporates physical ideas such as diffusion and advection [8]. This provides scientific support for the proposed wind-aware graph approach. Even though the proposed project may not fully implement a physics-guided differential equation model, the use of wind-based edge weighting follows the same physical intuition that pollutants move through the atmosphere rather than changing randomly.

2.5 Explainability in Air Quality Forecasting

Accuracy alone is not enough for an environmental decision-support system. If a model predicts that $PM_{2.5}$ will become high in a certain area, users should also understand why the model made that prediction. This is especially important in air quality forecasting because the prediction may influence public awareness, planning, and mitigation discussion.

GNNExplainer, proposed by Ying et al., provides a method for explaining predictions made by graph neural networks [9]. It identifies important parts of the graph and important node features that contribute to a prediction. In the proposed project, this idea can help identify which neighboring stations had the strongest influence on a forecasted pollution value.

However, station influence is only one part of the explanation. The system should also explain the role of meteorological and pollutant features such as wind speed, wind direction, humidity, temperature, and previous $PM_{2.5}$ values. SHAP, proposed by Lundberg and Lee, provides a general approach for explaining machine-learning predictions through feature contribution values [10]. This can be used to show which input features contributed most to a forecast.

By combining graph-based explanation and feature-level importance analysis, the proposed system can produce more meaningful outputs. Instead of only showing a forecasted $PM_{2.5}$ value, the system can explain that the prediction was influenced by rising $PM_{2.5}$ in previous hours, wind flow from a polluted neighboring station, low wind speed, or high humidity. This makes the model more understandable and useful for practical decision support.

2.6 Data Sources for Pollutant and Meteorological Information

The proposed system requires both air quality data and weather data. OpenAQ provides access to air quality measurements from monitoring stations and can be used to obtain pollutant data such as $PM_{2.5}$, PM_{10}, NO_2, and O_3 depending on station availability [11]. This is useful for building a reproducible data pipeline using publicly available air quality records.

Meteorological data is equally important because weather strongly affects pollutant concentration and movement. Open-Meteo provides historical weather data such as temperature, humidity, pressure, wind speed, wind direction, and wind gusts [12]. In the proposed project, wind speed and wind direction are especially important because they will be used to construct the wind-aware dynamic graph. Other weather variables can also help the model understand pollution accumulation and dispersion patterns.

2.7 Kathmandu Valley Air Quality Context

Since the project may later focus on Kathmandu Valley, it is important to understand the local air pollution context. Kathmandu Valley is a suitable case study because it has serious air pollution problems, high population density, rapid urbanization, and a valley-shaped geography that can affect pollutant accumulation. Becker et al. studied particulate matter variability in Kathmandu using in-situ measurements, remote sensing, and reanalysis data [13]. Their work is useful because it shows that Kathmandu's particulate matter pollution has strong temporal variability and is affected by seasonal and meteorological conditions.

Panday and Prinn studied the diurnal cycle of air pollution in Kathmandu Valley [14]. Their work is important because it shows that air pollution in Kathmandu changes throughout the day and is influenced by local meteorology and valley circulation. This supports the need for temporal modeling and wind-aware analysis in Kathmandu-focused air quality prediction.

Clean Energy Nepal also highlights that Kathmandu Valley is vulnerable to air pollution because of its bowl-shaped topography, limited wind flow, long dry season, rapid urbanization, and major emission sources such as transport, road dust, construction, brick kilns, household energy use, and waste burning [15]. This local context supports the motivation for developing a forecasting and visualization system for Kathmandu Valley.

ICIMOD reports that Nepal approved the Kathmandu Valley Air Quality Management Action Plan in 2020 and emphasizes the need for better air quality management and decision-support mechanisms [16]. This supports the practical relevance of the proposed dashboard and scenario-based analysis module. The proposed system will not replace official government warning systems, but it can serve as an academic research prototype for forecasting, explanation, visualization, and decision-support.

2.8 Research Gap

The reviewed literature shows that air quality forecasting has advanced significantly through deep learning and graph neural networks. Existing studies have shown that GNNs can model relationships between monitoring locations, domain knowledge can improve $PM_{2.5}$ prediction, and spatio-temporal models can capture both time-based and location-based pollution patterns. Studies such as PM2.5-GNN, HighAir, dynamic graph neural networks with adaptive edge attributes, and AirPhyNet provide strong support for graph-based, wind-aware, and physically meaningful air quality forecasting.

However, several gaps still remain. First, many forecasting systems use fixed graph structures, even though pollutant movement changes with wind direction and wind speed. Second, many models focus mainly on prediction accuracy but provide limited explanation of why a certain forecast was made. Third, existing systems often do not combine forecasting, explainability, risk visualization, and scenario-based decision support into one integrated dashboard. Finally, although Kathmandu Valley has been studied in terms of particulate matter variability, diurnal pollution cycles, and policy needs, there is limited work applying explainable wind-aware spatio-temporal graph neural networks to Kathmandu Valley air quality forecasting.

The proposed project addresses these gaps by developing an explainable wind-aware spatio-temporal graph neural network framework for air quality forecasting. The system will represent monitoring stations as graph nodes and construct dynamic edges using distance, pollutant correlation, wind speed, and wind direction. It will combine graph-based spatial learning with temporal models such as GRU or LSTM to forecast pollutant concentration. It will also include an explainability module to identify influential stations and important weather features. Finally, the results will be shown through an interactive dashboard for air quality visualization, risk analysis, and what-if scenario-based decision support.

3 METHODOLOGY

3.1 System Block Diagram/Architecture

The proposed system is organized into data collection, preprocessing, dynamic graph construction, spatio-temporal forecasting, explainability, what-if analysis, backend service, and dashboard layers.

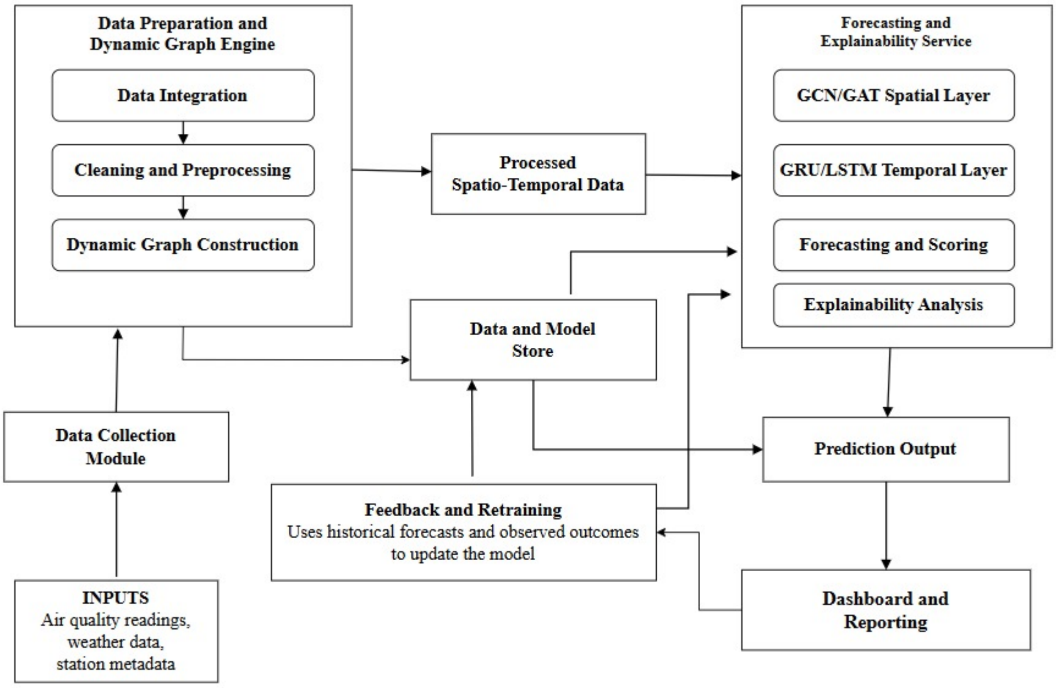

Figure 3-1 System Block Diagram

3.2 Description of Working Principle

The proposed system works as a wind-aware air quality forecasting and decision-support system. It combines air quality data, meteorological data, dynamic graph construction, spatio-temporal deep learning, explainability, and dashboard-based visualization. The overall working principle of the system is described in the following subsections.

3.2.1 Data Collection

The first step of the system is data collection from selected air quality monitoring stations and meteorological data sources. The air quality data consists of pollutant concentrations such as $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, and O_3. These pollutants are selected because they are commonly used indicators for assessing air pollution levels and public health risk. Along with pollutant values, station-related information such as station location, latitude, longitude, and timestamp is also collected.

In addition to air quality data, meteorological features are collected because weather conditions strongly influence the movement and concentration of pollutants. The meteorological data includes wind speed, wind direction, temperature, humidity, and atmospheric pressure. Among these features, wind speed and wind direction are especially important because they help determine how pollutants may travel from one monitoring station area to another. The collected data may be obtained from open environmental platforms, weather APIs, and publicly available benchmark datasets.

3.2.2 Data Cleaning and Preprocessing

After data collection, the raw data is passed through the preprocessing stage. Since air quality and meteorological datasets are collected from different sources, they may have different timestamp formats, missing readings, inconsistent sampling intervals, and abnormal values. Therefore, timestamp synchronization is performed to align pollutant and weather records according to a common time interval.

Missing values are handled using suitable methods such as interpolation, forward filling, or removal depending on the amount and pattern of missing data. Abnormal readings and outliers are also checked so that extremely incorrect sensor values do not affect the model training process. After cleaning, feature normalization or standardization is applied to scale the numerical values into a suitable range. This helps the forecasting model learn efficiently and prevents features with large numerical values from dominating other important features.

3.2.3 Dynamic Graph Construction

In the proposed system, each air quality monitoring station is represented as a node in a graph. The relationship between two stations is represented as an edge. Unlike a static graph, where station relationships remain fixed, this system uses a dynamic graph whose edge weights can change over time based on environmental conditions.

The edge weight between two stations is constructed using multiple factors. The first factor is spatial distance, where nearby stations are generally expected to have stronger influence on each other. The second factor is pollutant correlation, where stations showing similar pollution patterns are connected more strongly. The third and most important factor is wind information. If the wind direction indicates that pollutants can move from one station toward another, the connection between those stations becomes stronger. Wind speed is also considered because stronger wind may increase the movement of pollutants over a larger area.

By combining distance, pollutant similarity, wind direction, and wind speed, the system creates a wind-aware dynamic graph. This graph helps the model understand not only which stations are geographically close, but also which stations are likely to influence each other due to changing wind conditions.

3.2.4 Spatio-Temporal Forecasting Model

The processed data and the dynamic graph are given as input to the forecasting model. The forecasting model is designed to learn both spatial and temporal dependencies. Spatial dependency refers to the relationship between different monitoring stations, while temporal dependency refers to the change in pollutant concentration over time.

Graph-based neural network layers such as Graph Convolutional Network (GCN) or Graph Attention Network (GAT) are used to learn spatial relationships between stations. These layers use the dynamic graph to understand how pollution at one station may influence nearby or downwind stations. After spatial feature extraction, recurrent or sequence-based layers such as GRU or LSTM are used to learn temporal patterns from historical data. These temporal layers help the model understand how pollution levels change over hours or days.

The model is trained using historical air quality and weather data. During training, the predicted pollutant concentration is compared with the actual observed concentration. The prediction error is calculated using loss functions, and the model parameters are updated to reduce this error. After training, the model becomes capable of forecasting future pollutant concentrations for selected stations and time horizons.

3.2.5 Forecast Generation

Once the model is trained, it is used to generate future air quality forecasts. The system takes recent air quality readings, weather features, and current wind conditions as input. Based on these inputs, the dynamic graph is updated and the forecasting model predicts future pollutant concentrations.

The output may include predicted values of pollutants such as $PM_{2.5}$ or PM_{10} for the next few hours or days. These predicted values can also be converted into air quality risk levels such as good, moderate, unhealthy, or hazardous depending on the selected air quality index standard. This makes the output easier to understand for non-technical users.

3.2.6 Explainability Module

In addition to forecasting, the system includes an explainability module to make the model output more interpretable. Air quality forecasting models can be complex, especially when graph neural networks and temporal models are used. Therefore, explainability is important to understand why the model produces a certain forecast.

The explainability module identifies important weather features, influential neighboring stations, and significant time periods that contribute to the prediction. For example, the system may show that high $PM_{2.5}$ concentration at a station is influenced by strong wind flow from a nearby polluted station. Similarly, it may identify wind direction, humidity, or previous pollutant concentration as important factors for a high-pollution event. This helps users understand the reasoning behind the forecast rather than only viewing the final predicted value.

3.2.7 What-If Scenario Analysis

The system also includes a what-if scenario analysis module for decision support. This module allows users to observe how air quality may change under different hypothetical conditions. For example, users may analyze the possible effect of reduced emissions, changes in wind direction, or mitigation measures in a specific area.

By modifying selected input conditions and observing the forecasted output, the system can support environmental planning and policy-related analysis. Although the scenario results are not direct real-world guarantees, they provide useful insights into possible pollution trends and the relative impact of different environmental conditions.

3.2.8 Dashboard and Visualization

The final output of the system is presented through an interactive dashboard. The dashboard displays forecasted pollutant concentration, air quality risk levels, spatial pollution maps, trend graphs, and explanation-based insights. It allows users to view pollution patterns across different monitoring stations and understand how wind may affect pollutant movement.

The dashboard may also include reports and visual summaries for easier interpretation. Researchers can use the dashboard to analyze model behavior, while policymakers and concerned users can use it to understand air quality risk and possible mitigation strategies. Thus, the dashboard acts as the final decision-support interface of the proposed system.

3.2.9 Overall System Flow

Overall, the system starts by collecting air quality and meteorological data, followed by preprocessing and synchronization of the collected records. A dynamic graph is then constructed using station distance, pollutant correlation, and wind information. The spatio-temporal forecasting model uses this graph and historical data to predict future air quality. The explainability and scenario analysis modules provide meaningful interpretation of the results, and the final outputs are displayed through a dashboard for practical decision support.

3.3 Flowchart

Figure 3-2 presents the overall workflow of the proposed methodology.

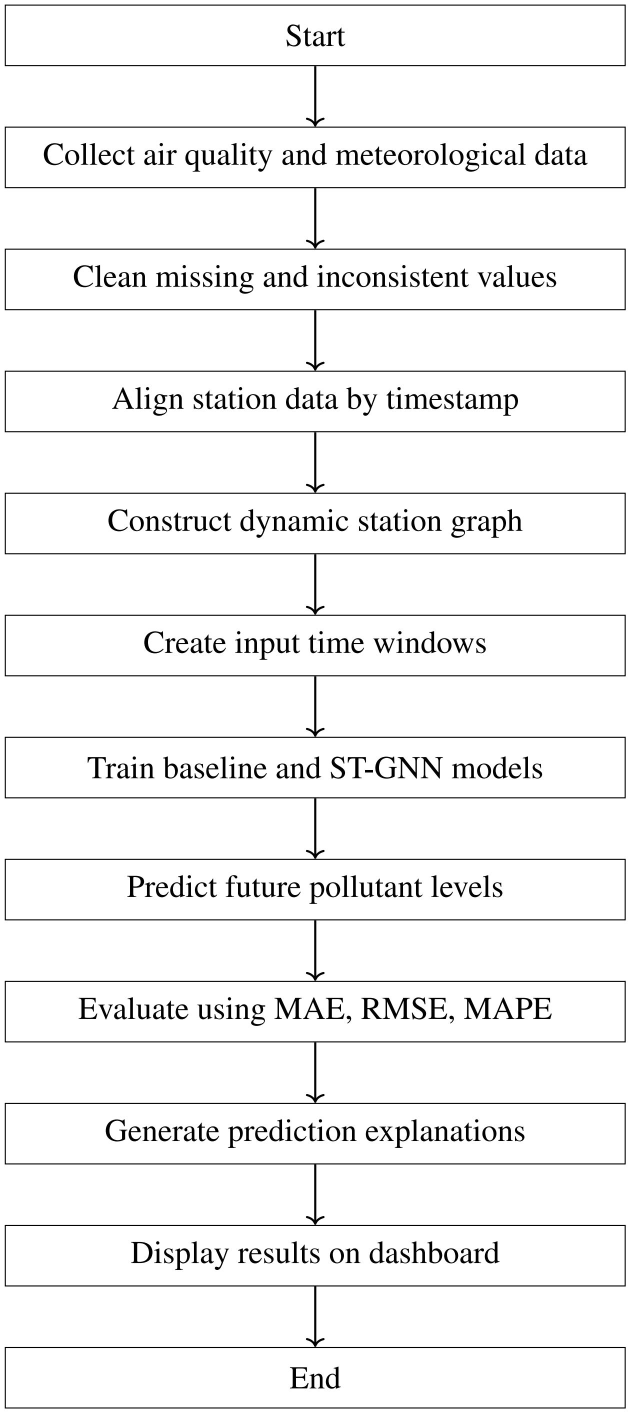

Figure 3-2 Proposed air quality forecasting workflow

3.4 Instrumentation Tools

Table 3-1 Proposed Technical Stack

Category	**Tools and Technologies**
Data Processing	Python, pandas, NumPy, scikit-learn
Deep Learning	PyTorch, PyTorch Geometric, optional Deep Graph Library (DGL)
Forecasting Models	Persistence baseline, regression, LSTM, GRU, GCN+GRU, GAT+GRU
Data Sources	OpenAQ, Open-Meteo, NASA POWER, public benchmark air quality datasets
Backend	FastAPI, Representational State Transfer (REST) services, experiment result APIs
Frontend	React or Next.js, Leaflet or Mapbox, Recharts
Database and Storage	PostgreSQL, local storage for datasets, models, and experiment logs
Deployment	Docker, local server, optional cloud deployment

3.5 Evaluation Methodology

The project will evaluate forecasting models using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), MAPE, and horizon-wise error analysis. The proposed model will be compared with baseline temporal models to check whether dynamic graph learning improves forecasting accuracy. Explainability will be assessed qualitatively by inspecting whether important stations, weather factors, and time windows are consistent with wind direction, distance, and pollutant history. The dashboard will be evaluated based on whether it can clearly present forecasts, risk levels, explanations, and what-if scenario comparisons.

4 EXPECTED OUTCOME

The primary goal of this project is to develop and validate a research-grade prototype for explainable, wind-aware air quality forecasting. The proposed system serves as a decision-support tool by bridging the gap between raw environmental data and actionable policy insights.

4.1 Technical Outcomes

The expected technical outcomes of the project are as follows:

- **Robust Data Pipeline:** A fully automated preprocessing framework for multi-source data ingestion, timestamp synchronization, outlier detection, missing value handling, and normalization to support high-quality model training.
- **Wind-Aware Dynamic Graph Engine:** A dynamic graph construction method that updates station-to-station relationships by integrating spatial distance, pollutant correlations, and wind direction and speed information.
- **Hybrid Forecasting Framework:** A forecasting framework consisting of traditional baseline models such as LSTM and GRU, along with advanced spatio-temporal graph neural network models such as GCN-GRU and GAT-GRU for performance benchmarking.
- **Explainability and Decision Support Module:** A module that interprets model outputs to provide human-readable insights, such as important meteorological features, influential neighboring stations, and temporal factors contributing to high-pollution events.
- **Interactive Analytics Dashboard:** A centralized visualization platform for displaying air quality forecasts, spatial risk maps, historical trends, and interpretability reports for end-users.

5 PROJECT SCHEDULE

The project is planned for approximately one academic year. Work will begin with literature review and dataset selection, followed by data collection, preprocessing, graph construction, baseline models, ST-GNN development, explainability, dashboard development, testing, documentation, and final presentation.

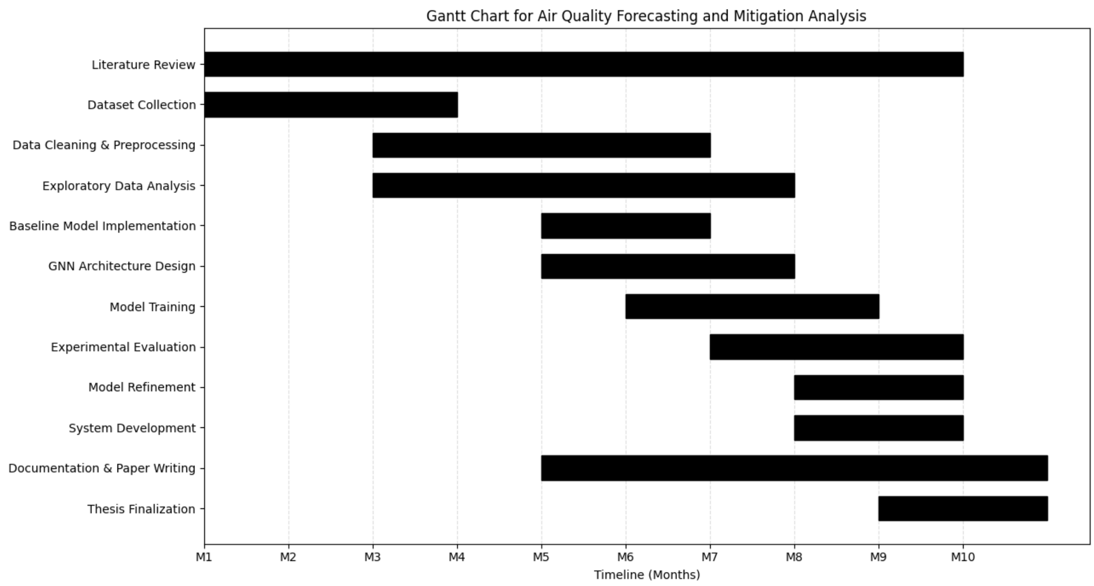

Figure 5-1 Gantt Chart

6 PROJECT BUDGET

The proposed system is mainly a software and research project, so the required budget is relatively low. Most libraries and datasets are open source or publicly available. The main expected expenses are internet access, optional cloud or Graphics Processing Unit (GPU) compute, deployment, documentation, and miscellaneous project needs.

Table 6-1 Estimated Project Budget

Category	**Description**	**Estimated Cost**
Internet and Data Access	Downloading datasets, accessing APIs, and reviewing research resources	NRs. 2,000
Cloud/GPU Usage	Optional Google Colab Pro, Kaggle, or equivalent compute for model training	NRs. 5,000
Domain/Hosting	Demo deployment for dashboard presentation, if needed	NRs. 3,000
Printing and Binding	Proposal, progress report, final report, and presentation materials	NRs. 3,000
Miscellaneous	Backup expenses and unforeseen project needs	NRs. 2,000
	Total Estimated Cost	**NRs. 15,000**

Most software tools used in this project will be open source. GPU usage may be reduced by using free Google Colab or Kaggle Notebook resources during early experiments.

7 FEASIBILITY ANALYSIS

7.1 Technical Feasibility

The project is technically feasible because public air quality datasets, weather APIs, and open-source machine learning frameworks are available. Data can be collected from OpenAQ, Open-Meteo, NASA POWER, or public benchmark datasets. The model can be developed using Python, PyTorch, PyTorch Geometric, pandas, and related tools. A smaller dataset or selected city can be used during the first phase to reduce computational complexity.

7.2 Operational Feasibility

The final system can be operated as a web-based dashboard. Users can view station-wise forecasts, risk levels, explanations, and what-if scenarios. The system is suitable as a research prototype and decision-support tool for academic demonstration. It is not intended to replace official government air quality warning systems.

7.3 Economic Feasibility

The project has low financial cost because most tools and datasets are open-source or publicly accessible. The main possible cost is cloud or GPU usage, which can be minimized using free resources such as Google Colab or Kaggle Notebook. The estimated budget is shown in Table 6-1.

7.4 Schedule Feasibility

The project is feasible within one academic year if developed in phases. The first working prototype can be built using baseline models and one selected dataset. The advanced ST-GNN, explainability, and scenario modules can be added progressively after the data pipeline and baseline models are stable.

7.5 Risk and Mitigation

Table 7-1 Project Risks and Mitigation Strategies

Risk	**Mitigation Strategy**
Missing or inconsistent sensor data	Use interpolation, timestamp alignment, quality checks, and selected stations with reliable coverage.
Limited station density	Start with a city or benchmark dataset that has enough monitoring stations for graph modeling.
Weather data mismatch	Use nearest weather grid points and align meteorological features by timestamp.
Long training time	Start with lightweight baseline models and use optional GPU resources only for selected experiments.
Overfitting	Use validation sets, early stopping, regularization, and comparison with simple baselines.
Weak explanation quality	Compare attention-based explanations with feature importance and physical wind-direction evidence.
Dashboard complexity	Build a simple MVP first, then add maps, scenario comparison, and reporting features.

7.6 Limitations

The system will not replace official air quality warning systems. Prediction accuracy will depend on data quality, station density, and availability of reliable weather data. The mitigation module will be treated as a what-if decision-support tool, not a guaranteed causal policy recommendation.

7.7 Final Feasibility Statement

The proposed project is feasible as a major project because the required data sources, algorithms, and development tools are available. It has sufficient research depth through dynamic graph construction, spatio-temporal forecasting, explainability, and mitigation scenario analysis while remaining practical enough to complete within the academic project period.

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